

Review

State-of-the-art one-stop handbook on wind forecasting technologies: An overview of classifications, methodologies, and analysis

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ABSTRACT

Wind energy has the inherent nature of intermittence and randomness, such that its accurate prediction is extremely critical to ensure safe and stable operation of power system with increased wind power integration. This paper aims to present a thorough state-of-the-art one-stop handbook on various approaches applied in wind forecasting based on three perspectives: wind speed and power forecasting, uncertainty forecasting, and ramp events forecasting. Firstly, four classifications of wind data according to data source, along with twenty-seven data pre-processing technologies which can efficiently improve prediction precision are carefully addressed. Then, a systematic literature review of model principle based wind speed and power forecasting strategies is investigated, which is categorized into three groups: physical approaches, statistical approaches, and combined approaches. Peculiarly, it has reported that combined approaches can realize more than 70% accuracy improvement compared with single model. Besides, three novel technologies developed in recent years are also discussed, e.g., spatial correlation forecasting, regional forecasting, and offshore forecasting. Moreover, thirty-seven evaluation criteria of wind speed and power forecasting are thoroughly summarized for performance verification. Additionally, uncertainty forecasting and ramp events forecasting which can provide more risk information for operators to handle decision-making issues in power system are also elaborated. Lastly, eight recommendations for further development in wind forecasting are also proposed. The prominent merit of this work is that a total of one hundred and seven wind forecasting methods from three perspectives are comprehensively summarized and compared based on inputs, time-scale, space-scale, forecasting variables, metrics, and features, which aims to help readers more effectively utilize these approaches for future in-depth researches.

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Nomenclature

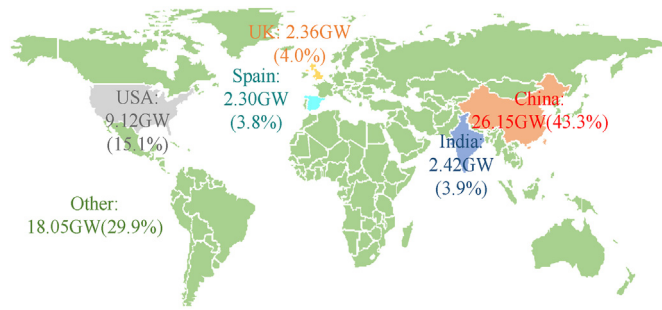
AE	Absolute error	MSE	Mean squared error
AR	Autoregressive model	NMAE	Normalized mean absolute error
ARCH	Autoregressive conditional heteroscedastic	NMAXAE	Normalized maximum absolute error
ARIMA	Autoregressive integrated moving average model	NNs	Neural networks
ARMA	Autoregressive moving average model	NRMSE	Normalized root mean square error
ASD	Atoms spare decomposition	NWP	Numerical weather prediction
CDF	Cumulative distribution function	PCA	Principal component analysis
CEEMD	Complementary ensemble empirical mode decomposition	PDF	Probability density function
CEEMDAN	Complete ensemble empirical mode decomposition with adaptive noise	PSR	Phase space reconstruction
DWT	Discrete wavelet transform	RE	Relative error
EEMD	Ensemble empirical mode decomposition	rMBE	Relative mean bias error
EMD	Empirical mode decomposition	RMSE	Root mean square error
EWT	Empirical wavelet transform	RMSRE	Root mean square relative error
FEEMD	Fast ensemble empirical mode decomposition	SDbias	Bias of standard deviation
ICEEMDAN	Improved complete ensemble empirical mode decomposition with adaptive noise	SDE	Standard deviation of errors
ITD	Intrinsic time-scale decomposition	SE	Sample entropy
KF	Kalman filter	SIA	Seasonal index adjustment
MAE	Mean absolute error	SS	Skill scores in terms of RMSE
MAPE	Mean absolute percentage error	SSA	Singular spectrum analysis
MASE	Mean absolute scale error	SSE	Sum of squared error
MPE	Mean percentage error	SSr	Skill scores in terms of RMSRE
mRMR	Maximum relevance minimum redundancy	SVMs	Support vector machines
		VMD	Variational mode decomposition
		WD	Wavelet decomposition
		WPD	Wavelet packet decomposition
		WTD	Wavelet threshold denoising

1. Introduction

In recent years, booming energy demand caused by rapid development of modern society (Yang et al., 2019a), severe environmental deterioration (Wang et al., 2020a), and shortage of fossil resources (Ming et al., 2020) are consistently threatening harmonious and healthy functioning of human society (Yang et al., 2019b). Fortunately, various renewable resources such as wind (Yang et al., 2016), solar (Chaibi et al., 2019), wave (Ulazia et al., 2019), biomass (Ishaq et al., 2020), geothermal (Li et al., 2020) and so on have

received ever-growing worldwide attentions and astonishing development over past decades (Li et al., 2019a), among which wind energy is envisaged as one of the most promising alternatives (Yang et al., 2018a). According to statistics of Global Wind Energy Council, increased global installed capacity of wind power in 2019 is more than 60.40 GW, while total capacity surpassed 651 GW (Global Wind Energy Council, 2019). Fig. 1 shows world's top five markets in 2019 and development trend of global wind power installed capacity in the last 20 years.

However, intermittency (Peng et al., 2020) and randomness



(a) Top five markets of wind power new capacity in 2019 (Total: 60.40 GW)



(b) Global wind power installed capacity progress in the last 20 years (GW)

Fig. 1. Development status of wind energy.

(Badal et al., 2019a) of wind speed tend to increase generation cost (He et al., 2020) and reduce power system reliability (Yang et al., 2018b), which hinder its further development and application (Song et al., 2020). Particularly, successive fluctuation of wind speed makes it difficult to figure out how much electricity will be injected into power grid (Li et al., 2020b), which might bring about energy transportation problems (Annamraju and Nandiraju, 2019). Therefore, accurate wind forecasting is absolutely required for wind power generation systems, which is crucial for facilitating wind energy penetration (Murty and Kumar, 2020), promoting wind energy consumption (Zafar et al., 2020), and maintaining power system stability (Yang et al., 2017).

Thus far, a large number of studies have been carried out to achieve a satisfactory wind forecasting performance, which can be divided into three groups: wind speed and power forecasting (Jung and Broadwater, 2014), uncertainty forecasting (Zhang et al., 2014), and ramp events forecasting (Cristobal et al., 2015). Note that input data of forecasting models are critical to final results precision under all three different classifications, while raw wind data always contains incorrect data and abnormal data resulted from unreliable sensors and unnatural operation states of generators (Zheng et al., 2015). Hence, numerous data pre-processing approaches are proposed to reduce forecasting errors, which are generally classified into decomposition-based methods (Qian et al., 2019), dimensionality deduction-based methods (Liu and Chen, 2019), and denoising-based methods (Mi et al., 2019). In particular, the former mainly includes wavelet decomposition (WD)-based algorithms (Cao and Li, 2008), empirical mode decomposition (EMD)-based algorithms (Wang et al., 2016), atoms sparse decomposition (ASD) (Cui et al., 2012), seasonal index adjustment (SIA) (Wang et al., 2015), intrinsic time-scale decomposition (ITD) (Zhang et al., 2016), and variational mode decomposition (VMD) (Naik et al., 2018). And feature selection (Vergara and Estévez, 2014) and feature extraction (Hira and Gillies, 2015) are two types of dimensionality deduction-based methods. Meanwhile, the last group mainly contains decomposition-based denoising methods (Liu and Chen, 2019), singular spectrum analysis (SSA) (Mi et al., 2019), and wavelet threshold denoising (WTD) (Cheng et al., 2018).

Meanwhile, early studies on wind speed and power forecasting can be traced back to 1970s (Wang et al., 2011), during which researches are focused on expected wind speed/power information prediction in next few minutes, hours, days, weeks, months or years. Until now, numerous devised wind speed and power forecasting approaches can be further classified according to time-scale (Wu and Hong, 2007), space-scale (Ma et al., 2009), model object (Soman et al., 2010), and model principle (Tascikaraoglu and Uzunoglu, 2014), while each classification also consists of several sub-classes (Hanifi et al., 2020). For instance, model principle based classification mainly contains three sub-groups: physical approaches, statistical approaches, and combined approaches. Note that physical approaches are also called numerical weather prediction (NWP) models (Lorenz, 1986), which heavily rely on complete descriptions of wind farm conditions (Bauer et al., 2015). However, statistical approaches put great attentions on time-varying relationships mining within time sequences, which are composed of three sub-categories: time series analysis methods, Kalman filter (KF), and machine learning methods (Lawan et al., 2014). Autoregressive model (AR) (Huang and Chalabi, 1995), moving average model (MA) (Huang and Chalabi, 1995), autoregressive moving average model (ARMA) (Torres et al., 2005), autoregressive integrated moving average model (ARIMA) (Yatiana et al., 2017), and autoregressive conditional heteroscedastic (ARCH) (Lv and Yue, 2011a) are typical time series analysis models, which often require high model orders to describe linear relationship of wind series. Meanwhile, dynamic propagation of model errors is considered in KF, which can effectively improve forecasting precision (Babazadeh et al., 2012). Besides, neural networks (NNs) (Azad et al., 2014) and support vector machines (SVMs) (Mohandes et al., 2004) are representative machine learning methods, which can establish nonlinear mapping between inputs and outputs by various learning rules to accurately describe wind randomness. Specially, NNs can be divided into traditional NNs and deep learning. The traditional NNs may fail to identify some complex features of wind, such as long-term dependency. Therefore, some deep learning methods have been introduced in wind forecasting field, e.g., convolutional neural network (CNN) (Zhao et al., 2020), recurrent neural network (RNN) (Cheng et al., 2018), long short-term memory (LSTM) (Liu et al., 2018a), neural network structural learning (Mi and Zhao, 2020), transfer learning (Hu et al., 2016), etc. Combined approaches integrating advantages of individual models are developed to achieve more accurate and reliable prediction. For example, weight coefficients assigning strategy is employed in horizontal combination methods (Han and Liu, 2010), while vertical combination approaches utilize different methodologies in different prediction links, e.g., main forecasting task, parameter selection and optimization, data pre-processing, and data post-processing (Liu and Chen, 2019). Specially, most of the previous work about combined approaches employ machine learning as main prediction algorithms. Then, meta-heuristic algorithms are adopted thanks to their merits of global optimization and rapid convergence, such as particle swarm optimization (PSO) (Kong et al., 2015), genetic algorithm (GA) (Santamaria-Bonfil et al., 2016), bat algorithm (BA) (Xiao et al., 2017), etc., which attempt to determine parameters to avoid time-consuming process. Meanwhile, data pre-processing approaches can be combined to filter out irrelevant and redundant features, improve the quality of input data, and decrease unnecessary computation burdens. In addition, residual error can be utilized in data post-processing to update prediction models for further prediction accuracy enhancement. Besides, novel technologies developed in last few years, including spatial correlation forecasting (Lin et al., 2017), regional forecasting (Lobo and Sanchez, 2012), and offshore forecasting (Perveen et al., 2014) have also attracted widespread attentions.

Table 1
Comparison of recent eight reviews.

Literature	Year	Highlights	Limitations
Wu (Wu and Hong, 2007)	2007	Presented a detailed literature review which focuses on advantages of each model.	Outdated technologies; Future works unstated.
Ma et al. (Ma et al., 2009)	2009	Provided a general survey on methods and developments of wind prediction.	Obscure or incomplete classifications; Lack of evaluation criteria.
Soman et al. (Soman et al., 2010)	2010	Reviewed wind forecasting approaches based on different time horizons.	Performance indicators, such as space-scale, forecast variables, and metrics are ignored.
Wang et al. (Wang et al., 2011)	2011	Analyzed accuracy of models and source of errors.	Many promising technologies and their especial application ignored.
Tascikaraoglu et al. (Tascikaraoglu and Uzunoglu, 2014)	2014	Reviewed combined methods on short-term wind prediction.	Incomplete classifications.
Jung et al. (Jung and Broadwater, 2014)	2014	Introduced a number of wind prediction technologies and forecasting accuracy.	Lack of detailed data pre-processing methods.
Zheng et al. (Qian et al., 2019)	2019	Discussed numerous decomposition-based hybrid models.	Incomplete classifications.
Hanifi et al. (Hanifi et al., 2020)	2020	Emphasized prediction time scale and accuracy improvement of each model.	Some promising technologies are ignored; Lack of detailed data pre-processing methods.

Generally speaking, wind forecasting accuracy is inevitably influenced by uncertain factors, such as model performance, mathematical equations, observation precision, and operational status of turbines (Song et al., 2018). Specifically, uncertain information greatly contributes to decision-making in power systems, such as trading strategy, unit commitment, and economic dispatch (Li et al., 2019b). Besides, uncertainty forecasting mainly includes three different categories, i.e., probabilistic forecasting (Yan et al., 2015), risk index forecasting (Pinson and Kariniotakis, 2004), and scenario forecasting (Pinson et al., 2009a).

Furthermore, wind power may sharply boost or drop under severe weather, which is known as ramp events (Cristobal et al., 2015). Obviously, ramp events can lead to active power imbalance, frequency stability deterioration, and even large-scale load shedding (Wang et al., 2017a). Hence, a series of studies on ramp events forecasting have been undertaken, which can be categorized into deterministic and probabilistic forecasting.

Until now, several reviews have been developed to discuss various wind forecasting strategies, which both have highlights and limitations, as can be summarized in Table 1.

To address the aforementioned drawbacks, this paper provides a systematic one-stop handbook about existing wind forecasting approaches based on state-of-the-art references, which aims to help readers more effectively utilize them for future in-depth researches. And the highlights/novelty of this work can be summarized as follows:

- Wind forecasting are reviewed based on three perspectives: wind speed and power forecasting, uncertainty forecasting, and ramp events forecasting;
- Three novel technologies developed in recent years, i.e., spatial correlation forecasting, regional forecasting, and offshore forecasting, are thoroughly discussed;
- Four classifications of wind data according to data source, along with twenty-seven data pre-processing technologies which can efficiently improve prediction precision are carefully addressed;
- Thirty-seven evaluation criteria are thoroughly summarized for performance verification;
- A total of one hundred and seven methods are comprehensively discussed based on inputs, time-scale, space-scale, forecasting variables, metrics, and features;
- All twenty-two sub-categories of wind forecasting approaches in three perspectives are carefully compared based on advantages/contributions, disadvantages/limitations, purposes, applicability, usage frequency, implementation difficulty, data size, and accuracy;
- Eight insightful recommendations for further development in wind forecasting are proposed.

Moreover, the structure of this paper is clearly shown in Fig. 2.

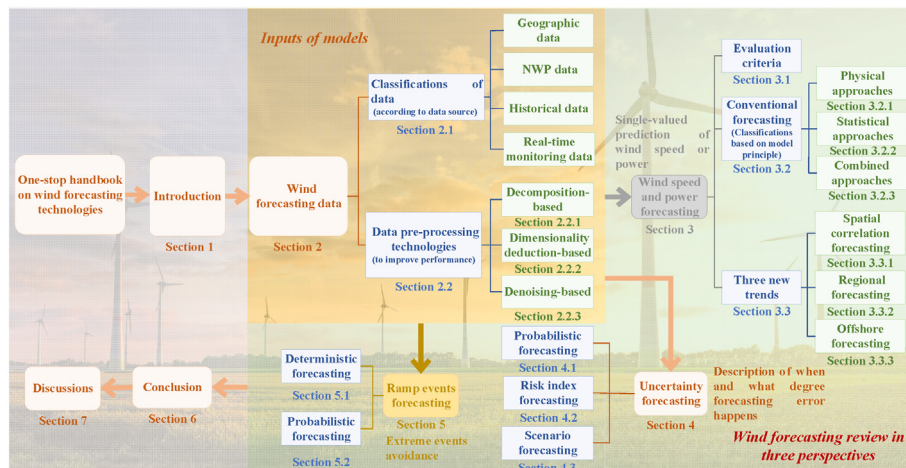


Fig. 2. Structure of this paper.

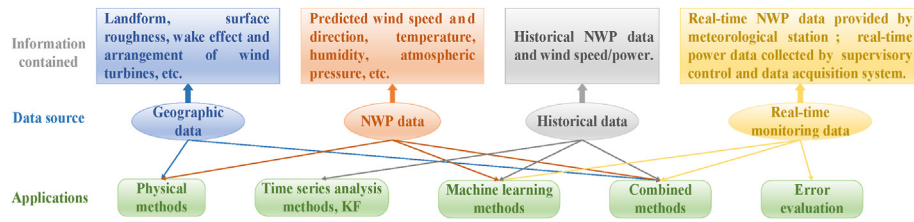


Fig. 3. Information contained in different data sources.

2. Wind forecasting data

2.1. Classifications of data

In general, wind forecasting data can be categorized into geographic data, NWP data, historical data, and real-time monitoring data according to data source, as summarized in Fig. 3. These data highly depend on complex measurement instruments located at adverse surroundings, which are associated with incorrect and abnormal data (Zheng et al., 2015). Hence, it is necessary to adopt data preprocessing to improve their prediction accuracy.

2.2. Data pre-processing

2.2.1. Decomposition-based approaches

These methods decompose unstable wind speed and power sequences into relatively stationary subsets. Then a prediction model on each subset is established to produce independent prediction outcomes such that the final result can be obtained by adding up all individual results (Liu and Chen, 2019). Note that they are mainly wavelet-based decomposition (Cao and Li, 2008), EMD-based decomposition (Wang et al., 2016), ASD (Cui et al., 2012), SIA (Wang et al., 2015), ITD (Zhang et al., 2016), and VMD (Naik et al., 2018), and. Particularly, WD-based decomposition methods are usually WD (Cao and Li, 2008), empirical wavelet transform (EWT) (Wang and Hu, 2015), wavelet packet decomposition (WPD) (Meng et al., 2016), and discrete wavelet transform (DWT) (Catalao et al., 2011). Besides, ensemble empirical mode decomposition (EEMD) (Santhosh et al., 2019), fast ensemble empirical mode decomposition (FEEMD) (Sun and Wang, 2018) complementary ensemble empirical mode decomposition (CEEMD) (Wang et al., 2017b), complete ensemble empirical mode decomposition with adaptive noise (CEEMDAN) (Zhang et al., 2017a), and improved complete ensemble empirical mode decomposition with adaptive noise (ICEEMDAN) (Song et al., 2018) are common variants of EMD.

2.2.2. Dimensionality deduction-based approaches

Dimensionality disaster might occur when there is too much data for decomposition (Liu and Chen, 2019). Feature selection and feature extraction are two groups to realize dimensionality reduction of data to improve accuracy by reducing computational complexity and redundant information. The difference is that feature selection returns a subset from existing features set, while feature extraction produces new features of original features. Feature selection approaches mainly include four categories, i.e., clustering-based methods (Gianniou et al., 2018), correlation-based methods (Zhao et al., 2020), information-based methods (Huang et al., 2018), and phase space reconstruction (PSR) (Sun and Wang, 2018). Particularly, clustering-based methods mainly are spectral clustering (SC) (Liu et al., 2015), k -means clustering (Wang et al., 2018), fuzzy c -means (FCM) clustering (Li et al., 2020c), kernel-based fuzzy c -means (KFCM) clustering (He et al., 2018), and self-organizing map (SOM) clustering (Wang et al., 2020b).

Autocorrelation function (ACF) (Zhao et al., 2020), partial correlation function (PACF) (Hu et al., 2018), and sample entropy (SE) (Sun and Wang, 2018) are common correlation-based methods. And information-based methods mostly include mutual information (MI) (Liu and Chen, 2019), conditional mutual information (CMI) (Huang et al., 2018) and mRMR (Xing et al., 2017). Moreover, principal component analysis (PCA) (Zhang et al., 2020) and Kernel principal component analysis (KPCA) (Sun et al., 2017) are frequently-used feature extraction approaches.

2.2.3. Denoising-based approaches

The aim of denoising is to remove noise information that affects prediction accuracy. Generally, decomposition-based denoising methods (Liu and Chen, 2019), SSA (Mi et al., 2019) as well as WTD (Cheng et al., 2018) are main denoising-based approaches in wind forecasting. Specially, decomposition-based denoising methods are associated with Section 2.2.1. Table 2 compares aforementioned twenty-seven data pre-processing approaches while a comprehensive comparison of three data pre-processing categories is listed in Table 3 (Liu and Chen, 2019). Note that no single method is absolutely superior to others. In fact, they are usually combined to further improve performance.

3. Wind speed and power forecasting

3.1. Evaluation criteria

To quantitatively evaluate performance of wind forecasting methods, thirty-seven evaluation criteria are summarized in Table 4.

3.2. Conventional forecasting approaches

Various classifications, e.g. time-scale, space-scale, and model object (Wang et al., 2011), are available for wind speed and power forecasting, as described in Fig. 4. Actually, wind speed forecasting and wind power forecasting have great similarities, thus they are reviewed together with an emphasis on model principle. And this section summarizes fifty-nine forecasting methods according to model principle based classification.

3.2.1. Physical approaches

Physical approaches are mainly based on NWP models, which utilize NWP data and geographic data to calculate wind speed at the height of wind turbine (Soman et al., 2010). Fig. 5(a) demonstrates their generalized flow chart. Physical approaches are greatly sensitive to initial conditions, which might result in deviations of wind speed values (Tascikaraoglu and Uzunoglu, 2014). Hence, they are often synthesized with other models to enhance forecasting precision (Hanifi et al., 2020).

3.2.2. Statistical approaches

Statistical approaches aim at establishing accurate mapping

Table 2
Chronological summary of twenty-seven data pre-processing approaches.

Approach			Year	Description	Advantages	Disadvantages
Decomposition-based	Wavelet-based	WD (Cao and Li, 2008)	2008	Decomposes original signals into different frequency bands via time domain and frequency domain transformation.	● Improved performance than traditional filters.	● Non-adaptive; Only detail signals decomposed; ● Rely on wavelet function and decomposition levels.
		DWT (Catalao et al., 2011)	2011		● Reduced redundant information.	● Non-adaptive; ● Rely on wavelet function and decomposition levels.
		EWT (Wang and Hu, 2015)	2015		● Adaptive; ● Independent on wavelet function and decomposition levels.	● Difficult to distinguish different time-varying components.
		WPD (Meng et al., 2016)	2016		● Both approximation and detail signals are decomposed.	● Non-adaptive; ● Rely on wavelet function and decomposition levels.
	EMD-based	EMD (Wang et al., 2016)	2012	Decomposes original signals into several intrinsic mode functions (IMFs) and one residue.	● Adaptive.	● Mode-mixing problem; ● Lack of strict mathematical theory.
		EEMD (Santhosh et al., 2019)	2016		● Adaptive; ● Mode-mixing problem solved.	● Poor accuracy due to residual; ● Lack of strict mathematical theory.
		CEEMD (Wang et al., 2017b)	2017		● Adaptive; ● Mode-mixing problem solved; ● Reconstruction precision.	● Illusive component problem; ● Lack of strict mathematical theory.
		CEEMDAN (Zhang et al., 2017a)	2017		● Adaptive; ● Mode-mixing problem solved; ● Reconstruction precision; ● Illusive component problem solved.	● Large computational resources; ● Spurious mode problem; ● Lack of strict mathematical theory.
		FEEMD (Sun and Wang, 2018)	2018		● Adaptive; ● Mode-mixing problem solved; ● Fast calculation.	● Sensitive to noises and sampling; ● Lack of strict mathematical theory.
		ICEEMDAN (Song et al., 2018)	2018		● Adaptive; ● Mode-mixing problem is solved; ● Reconstruction precision; ● Illusive component problem is solved; ● Reduced residual noise.	● Lack of strict mathematical theory.
		ASD (Cui et al., 2012)	2012	Decomposes original signals into a group of optimal atoms in highly redundant atomic library.	● Greedy self-adaptive search; ● Sparse result.	● Abundant atoms in redundant library; ● Heavy calculation burden.
		SIA (Wang et al., 2015)	2015	Decomposes original signals into seasonal and trend components.	● Seasonal factors are adopted to handle nonlinear signals.	● Difficult to determine which modelling approach (multiplication or addition) is appropriate.
		ITD (Zhang et al., 2016)	2016	Decomposes original signals into a group of proper rotation components and one monotonous trend.	● Mode-mixing problem solved; ● Especially suitable for nonlinear and non-stationary signals.	● Lack of strict mathematical theory.
		VMD (Naik et al., 2018)	2018	Decomposes original signals into a series of modes with different center frequencies.	● Non-recursive variational model; ● Solid mathematical theory; ● Strong robustness.	● Difficult to distinguish different time-varying components.
Dimensionality deduction-based	Feature selection	Clustering (Gianniou et al., 2018)	2015	Divides original signals into several clusters, then selects the most similar clusters with forecast target while abandoning valueless ones.	● Reduced calculation burden.	● Redundant information included.
		Correlation-based SE (Sun and Wang, 2018)	2018	Evaluates complexity of sequences.	● Independent data length;	● Low robustness.

Table 2 (continued)

Approach	Year	Description	Advantages	Disadvantages
Information-based	PACF (Hu et al., 2018)	2018 Reveals correlation between features.	● Relatively trouble-free implementation.	● More computation time than ACF.
	ACF (Zhao et al., 2020)	2020	● Correlation of residuals with next lagged value provided.	● Interference information included.
	mRMR (Ma et al., 2009)	2017 Obtains a maximally relevant and minimally redundant set based on mutual information.	● Correlation of series with its lagged values provided.	● Single redundancy and relevance evaluation.
	MI (Huang et al., 2018)	2018 Measures amount of information that one feature has about another feature.	● Maximum relevant components with minimum redundancy.	● Redundancy inside features ignored.
	CMI (Huang et al., 2018)	2018	● Relevance between features analyzed.	● Complex mathematical theories.
	PSR (Sun and Wang, 2018)	2018 Analyses dynamic characteristics of time and mapping series into phase space.	● Both relevance and redundancy are analyzed between features.	● Rely on selection of embedding dimension and delay time
	KPCA (Sun et al., 2017)	2017 Reduces dimensionality by maximizing the variance of each dimension.	● Suitable for unstable sequences.	● Rely on rigorous mathematical derivation.
	PCA (Zhang et al., 2020)	2020	● Nonlinear.	● Rely on rigorous mathematical derivation;
			● Simple.	● Linear.
			● Simple and less computation time.	● Rely on wavelet function and decomposition levels.
Denoising-based	WTD (Cheng et al., 2018)	2018 Obtains several subseries, then wavelet coefficients of noises at each scale are discarded.	● Solid mathematical theory;	● Phase shift phenomenon.
	SSA (Mi et al., 2019)	2019 Identifies noise, trend, periodic oscillation of original signals.	● Few parameters.	
	Decomposition-based denoising (Liu and Chen, 2019)	2019 Obtains several subseries, then some of these are selected as noises and removed.	● Clear framework.	● Prediction accuracy might be regraded.

Table 3

A comprehensive comparison of three data pre-processing categories.

Approach	Purposes	Results	Applicability	Usage frequency	Implementation difficulty	Robustness	Computation time	Accuracy improvement
Decomposition-based	Reduce prediction difficulty.	Relatively stationary subseries.	Medium or small data with relatively low fluctuation.	****	****	*	****	****
Dimensionality deduction-based	Feature selection	Select features with high relevance.	Subset of initial sets.	Large data with high relevance.	***	*	**	***
	Feature extraction	Establish reduced feature sets with removing redundancy.	New feature sets.	Large data with high redundancy.	**	**	***	**
Denoising-based	Strengthen validity of input data.	Denoised subseries.	Medium data with drastic fluctuation.	*	***	****	*	***

*Note. *: very low/short; **: low/short; ***: high/long; ****: very high/long.

between inputs (i.e., NWP data, historical data, geographic data, etc.) and outputs (wind speed or power), which mainly consist of time series analysis approaches, KF, and machine learning approaches, as shown in Fig. 4(b). Specifically, time series analysis approaches can predict wind speed in next few minutes based on historical data without an accurate system model (Bossanyi et al., 1980). KF is a statistically optimal sequential estimation procedure for dynamic systems, which relies on a series of mathematical calculations (Tian et al., 2014). In KF, wind observations are recursively combined with recent forecasts using weights to minimize corresponding biases (Babazadeh et al., 2012). Besides, neural networks (NNs) (Li and Shi, 2010) and SVMs (Maria et al., 2014) can build inductive model between inputs and outputs based on various learning rules without detailed system information.

Particularly, Table 5 summarizes twenty-five statistical approaches in publications. In conclusion, conventional statistical methods (time series analysis approaches and KF) are easy to implement and mostly applied to very short-term and short-term forecasting, which are usually regarded as reference models. While machine learning approaches have a wider applicability, i.e., all time-term, but highly rely on selected parameters. Hence, these approaches are often combined with parameter selection and optimization algorithms, data pre-processing methods, and data post-processing methods to construct combined models.

3.2.3. Combined approaches

Various methods both have their unique merits and inevitable shortcomings, which motivate combined approaches to achieve

Table 4
Thirty-seven evaluation criteria of wind forecasting.

Categories	Criteria	Expression	Purpose	Applicability	Usage frequency	Complexity
Fundamental metrics	Absolute error (AE) (Lydia et al., 2014)	$ p_e^i - p_a^i $	Describes difference between estimated and actual value.	Single object evaluation.	**	*
	Bias (Wang et al., 2019)	$\frac{1}{N} \sum_{i=1}^N \left(\frac{p_e^i - p_a^i}{p_{\text{Nom}}^i} \right)$	Dimensionless description of deviation.	Dimensionless multiple objects evaluation.	****	**
	Bias of standard deviation (SDbias) (Wang et al., 2019)	$\text{sd}(p_e^i) - \text{sd}(p_a^i)$	Measures standard deviation between expected value and true value.	Multiple objects evaluation with large difference.	**	***
	Relative error (RE) (Lydia et al., 2014)	$\left \frac{p_e^i - p_a^i}{p_a^i} \right \times 100\%$	Describes reliability of predicted results.	Multiple objects evaluation with minor difference.	**	*
Mean value-based metrics	Mean absolute error (MAE) (Lydia et al., 2014)	$\frac{1}{N} \sum_{i=1}^N p_e^i - p_a^i $	Reflects overall level of errors.	Multiple objects evaluation with large-scale data.	*****	**
	Mean absolute percentage error (MAPE) (Lydia et al., 2014)	$\frac{1}{N} \sum_{i=1}^N \left \frac{p_e^i - p_a^i}{p_a^i} \right \times 100\%$	Intuitive interpretation of relative error.	Intuitive and uniform representation required.	*****	***
	Mean absolute relative error (MARE) (Velazquez et al., 2011)	$\frac{1}{N} \sum_{i=1}^N \left \frac{p_e^i - p_a^i}{p_a^i} \right $	Reflects average absolute value of relative differences between estimated and actual value.	Multiple objects evaluation with medium-scale data.	*	***
	Mean absolute range normalized error (MARNE) (Panapakidis et al., 2019)	$\frac{1}{N} \sum_{i=1}^N \left \frac{p_e^i - p_a^i}{\max(p_a^i)} \right \times 100\%$	Describes absolute difference between estimated and actual value, normalized to maximum wind speed/power.	Intuitive and clear representation required.	*	****
	Mean absolute scale error (MASE) (Peng et al., 2017a)	$\frac{(N-1) \sum_{i=1}^N p_e^i - p_a^i }{N \sum_{i=2}^N p_e^i - p_a^{i-1} }$	Determines comparative accuracy of prediction.	High precision required.	**	*****
	Mean percentage error (MPE) (Wang et al., 2015)	$\frac{p_a^i - p_e^i}{p_a^i} \times 100\%$	Represents average of percentage errors between estimated and actual value.	Intuitive and uniform representation required.	**	**
	Mean relative error (MRE) (Wang et al., 2019)	$\frac{1}{N} \sum_{i=1}^N \left \frac{p_e^i - p_a^i}{p_a^i} \right $	Approximation error measure based on mean of relative errors	Multiple objects evaluation with medium-scale data.	*	***
	Mean squared error (MSE) (He et al., 2018)	$\frac{1}{N} \sum_{i=1}^N (p_e^i - p_a^i)^2$	Reveals contribution of positive and negative errors to lack of accuracy.	Multiple objects evaluation with minor difference.	**	**
	Relative mean bias error (rMBE) (Zhang et al., 2020)	$\frac{1}{N} \sum_{i=1}^N \frac{p_a^i - p_e^i}{p_a^i}$	Assesses forecasting bias.	Multiple objects evaluation with large-scale data.	*	**
Normalized value-based metrics	Normalized mean absolute error (NMAE) (Wang et al., 2019)	$\frac{1}{N} \sum_{i=1}^N \left \frac{p_e^i - p_a^i}{p_{\text{Nom}}^i} \right $	Dimensionless comparison of overall level for errors.	Dimensionless multiple objects evaluation with large-scale data.	***	**
	Normalized maximum absolute error (NMAXAE) (Zhang et al., 2016)	$\max \left(\left \frac{p_e^i - p_a^i}{p_{\text{Nom}}^i} \right \right) \times 100\%$	Dimensionless description of maximum deviation.	Intuitive and uniform comparison required.	**	**
	Normalized mean square error (NMSE) (Wang et al., 2017b)	$\frac{1}{N} \sum_{i=1}^N \left(\frac{p_e^i - p_a^i}{\text{Cap}} \right)^2$	Dimensionless description of MSE.	Dimensionless multiple objects evaluation with minor difference.	***	***
	Normalized root mean square error (NRMSE) (Wang et al., 2019)	$\sqrt{\frac{1}{N} \sum_{i=1}^N \left(\frac{p_e^i - p_a^i}{p_{\text{Nom}}^i} \right)^2}$	Dimensionless description of RMSE.	Dimensionless multiple objects evaluation with minor difference.	*****	***
Root mean square-based metrics	Root mean square error (RMSE) (Zhao et al., 2011a)	$\sqrt{\frac{1}{N} \sum_{i=1}^N (p_e^i - p_a^i)^2}$	Describes aggregate magnitudes of errors for various times into a single metric.	Multiple objects evaluation with minor difference.	*****	***
	Root mean square relative error (RMSRE) (Wang et al., 2019)	$\sqrt{\frac{1}{N} \sum_{i=1}^N \left(\frac{p_e^i - p_a^i}{p_a^i} \right)^2}$	Assesses deviation from actual value.	Multiple objects evaluation with tiny difference.	****	***
Coefficient	Correlation coefficient (r) (Wang et al., 2017b)	$\frac{\sum_{i=1}^N (p_a^i - \bar{p}_a)(p_e^i - \bar{p}_e)}{\sqrt{\sum_{i=1}^N (p_a^i - \bar{p}_a)^2} \sqrt{\sum_{i=1}^N (p_e^i - \bar{p}_e)^2}}$	Represents linear correlation between forecasted and actual value.	Medium-scale data.	***	*****
	Coefficient of determination (R^2) (Lydia et al., 2014)	$1 - \frac{\sum_{i=1}^N (p_e^i - p_a^i)^2}{\sum_{i=1}^N (p_a^i - p_{a,m}^i)^2}$	Describes how well model fits data.	Large-scale data.	****	***
	Directional change of the forecasting results (DC) (Wang et al., 2017b)	$\frac{1}{N} \sum_{i=1}^N a_i$ $a_i = \begin{cases} 1, & \text{if } (p_e^{i+1} - p_e^i)(p_a^{i+1} - p_a^i) > 0 \\ 0, & \text{otherwise} \end{cases}$	Reflects prediction movement directions or turning points.	Medium-scale data.	**	***
				Small-scale data.	**	****

Table 4 (continued)

Categories	Criteria	Expression	Purpose	Applicability	Usage frequency	Complexity
Other metrics	Index of agreement of forecasting results (IA) (Wang et al., 2017b)	$1 - \frac{\sum_{i=1}^N (P_e^i - P_a^i)^2}{\sum_{i=1}^N (P_e^i - \bar{P}_a + P_a^i - \bar{P}_a)^2}$	Detects additive and proportional differences in estimated and actual means.			
	Skill scores in terms of RMSE (SS) (Wang et al., 2019)	$\frac{\text{NRMSE}(\Phi_{\text{fit}}) - \text{NRMSE}(\Phi_m)}{\text{NRMSE}(\Phi_l) - \text{NRMSE}(\Phi_m)}$	Measures effectiveness of fitting.	Small-scale data.	*	****
	Skill scores in terms of RMSRE (SS _r) (Wang et al., 2019)	$\frac{\text{RMSRE}(\Phi_{\text{fit}}) - \text{RMSRE}(\Phi_m)}{\text{RMSRE}(\Phi_l) - \text{RMSRE}(\Phi_m)}$	Measures effectiveness of fitting.	Small-scale data.	*	****
	Median of <i>N</i> absolute percentage error (MdAPE) (Wang et al., 2017b)	$\text{median}\left(\left \frac{P_e^i - P_a^i}{P_a^i}\right \times 100\%\right)$	Compares forecasting methods across different series.	Intuitive and uniform representation required.	*	***
	Standard deviation of errors (SDE) (Wang et al., 2019)	$\sqrt{\frac{1}{N} \sum_{i=1}^N \left[\left(\frac{P_e^i - P_a^i}{P_{\text{Nom}}^i} \right) - \text{Bias} \right]^2}$	Measures dispersion of data in relation to mean.	Dimensionless multiple objects evaluation with small-scale data.	***	*****
Uncertainty forecasting metrics	Sum of squared error (SSE) (Wang et al., 2019)	$\sum_{i=1}^N (P_e^i - P_a^i)^2$	Measures fitting performance.	Multiple objects evaluation with minor difference.	***	**
	Average coverage error (ACE) (Wang et al., 2017a)	PICP – PICN	Describes reliability of predictive intervals.	Small-scale data.	**	****
	Coverage width-based criterion (CWC) (Wang et al., 2017a)	$\text{PINAW}[1 + I(\text{PICP})e^{-\rho(\text{PICP}-v)}], \rho \in [50, 100]$ $I(\text{PICP}) = \begin{cases} 1, & \text{if } \text{PICP} < v \\ 0, & \text{otherwise} \end{cases}$	Comprehensively evaluates quality of constructed predictive intervals.	Small-scale data.	*	*****
	Kurtosis (Zuluaga et al., 2015)	$\left\{ \frac{N(N-1)}{(N-1)(N-2)(N-3)} \sum_{i=1}^N \left(\frac{P_e^i - P_a^i}{\text{SD}} \right)^4 \right\} \times \frac{3(N-1)^2}{(M-2)(M-3)}$	Explains degree of distribution peak.	Small-scale data.	***	*****
	Prediction interval coverage probability (PICP) (Wang et al., 2017a)	$\text{PICP} = \frac{1}{N} \sum_{i=1}^N x_i, x_i = \begin{cases} 1, & \text{if } L_i < P_a^i < U_i \\ 0, & \text{otherwise} \end{cases}$	Measures reliability of predictive intervals.	Medium-scale data.	****	***
	Prediction intervals normalized average width (PINAW) (Wang et al., 2017a)	$\frac{1}{N(\max_p - \min_p)} \sum_{i=1}^N U_i - L_i$	Measures width of prediction intervals.	Large-scale data.	**	**
	Skewness (Zuluaga et al., 2015)	$\frac{N}{(N-1)(N-2)} \sum_{i=1}^N \left(\frac{P_e^i - P_a^i}{\text{SD}} \right)^3$	Estimates symmetry of distribution.	Medium-scale data.	***	*****
Scenario forecasting metrics	Maximum deviation (Sumaili et al., 2011)	$\frac{1}{N} \sum_{i=1}^N P_e^i - \max(P_e^i) $	Measures maximum deviation between estimated and actual value.	Low precision required.	*	**
	Mean deviation (Morales et al., 2010)	$\frac{1}{N} \sum_{i=1}^N P_e^i - \bar{P}_e $	Measures mean deviation between estimated and actual value.	Rapid comparison.	***	**
	Standard deviation (SD) (Morales et al., 2010)	$\sqrt{\frac{1}{N} \sum_{i=1}^N (P_e^i - \bar{P}_e)^2}$	Measures standard deviation between estimated and actual value.	Multiple objects evaluation with minor difference.	***	***

Note.** Φ_{fit} : fitting power curves; Φ_l : ideal power curves; Φ_m : manufacturer's power curves; C_{ap} : capacity of wind energy; L_i : lower bound of P_i ; $\max_p$: maximum forecasting target; $\min_p$: minimum forecasting target; N : length of data set; P_a^i : the *i*th actual power data; P_e^i : the *i*th estimated power data; $\bar{P}_a$: mean of actual power; $\bar{P}_e$: mean of estimated power; $P_{a,m}^i$: mean of the actual power; P_{Nom}^i : nominal power; PINC: prediction interval nominal confidence which is 10–90%; $r(\cdot)$: correlation coefficient between estimated power and actual power; $\text{sd}(\cdot)$: standard deviation between the estimated power and actual power; U_i : upper bound of P_i ; v : nominal confidence level; *: very low; **: low; ***: medium; *: high; *****: very high.

more satisfactory forecasting performance compared to individual method.

Forecasting outcome of horizontal combination approach is generated via assigning weight coefficients to respective result of each single method, as depicted in Fig. 6. Undoubtedly, determination of weight coefficients is crucial to forecasting performance. Particularly, entropy principle (Han and Liu, 2010), Gaussian process regression (GPR) (Wang and Hu, 2015), support vector regression (SVR) (Li et al., 2018), Bayesian model (Wang et al., 2020b), equal weight average method (Zhao et al., 2011b), and covariance optimization combination method (Zhao et al., 2011b) are some representative technologies. Meanwhile, meta-heuristic algorithms such as modified flower pollination algorithm (MFPA)

(Zhang et al., 2017a), grey wolf optimizer (GWO) (Song et al., 2018), and multi-objective grasshopper optimization algorithm (MOGOA) (Niu and Wang, 2019) are employed to acquire weight coefficients, which own merits of rapid convergence and global optimization (Zhang et al., 2017a). In addition, vertical combination approaches often adopt different methodologies in different prediction tasks, together with auxiliary processes: parameter selection and optimization, data pre-processing, and data post-processing. It is reported that certain parameters of models make a great influence on forecasting results (Rubio et al., 2011). Parameter selection and optimization methods also utilize meta-heuristic algorithms to obtain optimal parameters. Data pre-processing methods can improve overall data quality to avoid unnecessary computation



Fig. 4. Classifications of wind speed/power forecasting considering time-scale, space-scale, and model object.

burdens by removing redundant data characteristics (Bokde et al., 2019). Data post-processing methods focus on decreasing negative effects of systematic errors. In these models, the final prediction models are revised by residual error prediction models, which have three steps: (i) establish preliminary prediction model and calculate residual errors; (ii) establish residual errors prediction models; (iii) update elementary prediction model by forecasting residual errors and acquire final results (Davò et al., 2016). Fig. 7 illustrates their generalized flowchart.

Moreover, Table 6 offers forty-one state-of-the-art combination methods in the last ten years. In theory, an increase of combined methods will result in a growth of precision improvement, as well as longer time. Hence, how to realize a proper trade-off between precision and efficiency is one of most important research directions.

3.3. Three new trends

3.3.1. Spatial correlation forecasting

Spatial correlation forecasting is widely used for assessing wind resources at site where information is unavailable but neighbouring measuring stations are available (Hu et al., 2016). Physical approaches derive wind speed relationship among each site according to topographic factors (e.g., temperature, humidity, pressure, etc.) and spatial topologies (e.g., landform, surface roughness, wake effect, arrangement of wind turbines, etc.) (Chan et al., 1983). On the other hand, statistical approaches such as artificial NNs (Velazquez et al., 2011), fuzzy model (Damousis et al., 2004), and measure-correlate-predict (Ali et al., 2018) can directly establish mapping of wind speed/power via analyzing historical samples at different spatial sites.

3.3.2. Regional forecasting

Spatial smoothing effects imply that whole forecasting errors will shrink as regional scale expands (Santamaría-Bonfil et al., 2016). Hence, regional forecasting can provide more accurate information for operators, while comparison of common three kinds regional forecasting methods is demonstrated in Fig. 8.

Moreover, a summary of four spatial correlation forecasting methods and six regional forecasting methods in the last fifteen years are illustrated in Table 7. Compared with conventional wind speed and power forecasting, spatial correlation forecasting needs more measured data from multiple spatial correlated sites which cause high complexity and cost. Superposition method of regional forecasting is rarely employed due to its heavy computation burden. Statistical upscaling method is a mature and widely used method so far although its performance relies on selected reference wind farms. Lastly, SRMA is a recent super-competitive method, which owns a higher accuracy and less computation time than that of statistical upscaling method.

3.3.3. Offshore forecasting

Offshore wind farms are generally more attractive than onshore wind farms, which have the unique merits of stronger and more constant wind speed, as well as greater installed capacity. Here, differences between offshore and onshore wind forecasting can be outlined as follows:

- From the perspective of measurement equipment. Offshore wind forecasting depends on sensors located at harsh ocean environment. Undoubtedly, offshore sensors are susceptible to damage

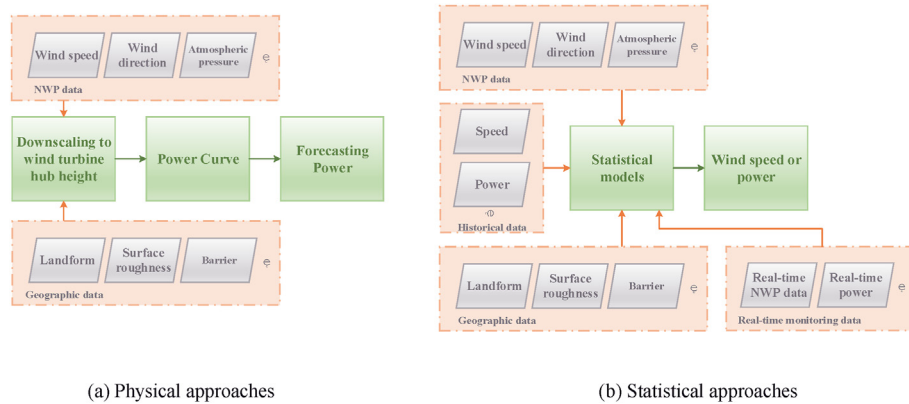


Fig. 5. Generalized flow chart of physical approaches and statistical approaches.

Table 5
Chronological summary of twenty-five statistical methods for wind speed and power prediction.

Approach		Year	Inputs	Time-scale	Space-scale	Forecast variables	Metrics	Features
Time series analysis	Persistence method (Bossanyi et al., 1980)	1980	V_h	Short	Single turbine	V	/	● Benchmark model.
	AR (Huang and Chalabi, 1995)	1995	V_h	Short	Single turbine	V	R^2	● Time-varying, low-order, and linear model.
	ARMA (Torres et al., 2005)	2005	V_h	Short	Single turbine	V	MAE, RMSE	● RMSE less than 1.5 m/s.
	Fractional-ARIMA (Kavasseri and Seetharaman, 2009)	2009	V_h	Short	Single wind farm	P, V	$\sqrt{SSE}$, daily mean error, variance	● Improved 42% precision than persistence method; ● Limitation of cut-in and cut-out wind speed.
	GARCH (Zhou et al., 2010)	2010	V_h	Short	Single turbine	P	Daily average relative error	● Suitable for sequences with greater volatility clustering.
	ARIMA-ARCH (Lv and Yue, 2011b)	2011	V_h	Short	Single wind farm	V	MRE	● Heteroscedasticity effect considered.
	GARCH-M (Chen et al., 2013)	2013	P_h	Short	Single wind farm	P	MAE, MAPE, RMSE	● Volatility and intermittency of wind described.
	CGARCH-M (Chen et al., 2015)	2015	P_h	Short	Single wind farm	P	MAE, MAPE, RMSE	● Permanent and transitory components of data considered.
KF	ARIMA (Yatiyana et al., 2017)	2017	V_h	Short	Single turbine, Single wind farm	V	MAPE	● Forecasting error less than 5% for wind speed.
	Kalman filter (based on Gauss-Markov) (Babazadeh et al., 2012)	2012	V_h	Very short, Short	Single wind farm	V	RMS	● Suitable for an hour ahead prediction.
	Adaptive KF (Tian et al., 2014)	2014	V_h	Short	Single wind farm	V	MAPE	● Prediction error and time delay compensated; ● Easy to derive Kalman state and measurement equations.
	Unscented KF (Chen and Yu, 2014)	2014	V_h	Short	Single turbine	V	MAPE, R^2 , RMSE	● Strong reliability and robustness; ● Recursively updated.
	Weighted robust KF (Zuluaga et al., 2015)	2015	V_h	Very short	Single turbine	V	Kurtosis, Skewness	● Weighted robust KF and robust statistics KF can obtain better performance than thresholded KF.
	Robust statistics KF (Zuluaga et al., 2015)							
	Thresholded KF (Zuluaga et al., 2015)							
	Feed-forward BP (Li and Shi, 2010)	2010	V_h	Short	Single turbine	V	MAE, MAPE, RMSE	● No single NN model is superior to other models in all evaluation indicators; ● Performance depends on inputs data.
Machine learning	Radial basis function (Li and Shi, 2010)							
	ADALINE NNs (Li and Shi, 2010)							
	NNs (based on data fusion algorithm) (Azad et al., 2014)	2014	V_h	Long	Single turbine	V	MAE	● Long-term mean hourly prediction.
	Transfer learning (Hu et al., 2016)	2016	V_h	Very short, Short	Single wind farm	V	MAE, MAPE, MSE, RMSE	● Satisfactory prediction precision; ● Requirement of sufficient training data.
	ANNs (Filik and Filik, 2017)	2017	V_h , NWP data	Very short	Single turbine	V	MAE, RMSE	● Multiple input variables utilized.
	SVMs SVM (Gaussian kernel) (Mohandes et al., 2004)	2004	V_h	Long	/	V	MSE	● Higher SVM order (from 1 to 11) improves forecasting accuracy.
	Least-squares SVM (Maria et al., 2014)	2014	V_h , NWP data	Short	Single wind farm	P	Bias, NMAE, RMSE	● Excellent precision and generalization capability; ● Fewer parameters and shorter computational time.
	SVM-enhanced Markov model (Yang et al., 2015)	2015	V_h	Very short, Short	Single wind farm	P	MAE, RMSE	● Diurnal non-stationarity and seasonality of wind power as well as wind ramps considered.
	SVM based on multi-variable regression (Park and Hur, 2017)	2017	V_h	Short	Single turbine	P	RMSE	● Forecasting accuracy guaranteed by analyzing reliability of power curves.

*Note. ADALINE: adaptive linear element; ANNs: artificial NNs; BP: back propagation; CGARCH-M: component generalized autoregressive conditional heteroskedasticity in mean; GARCH: generalized autoregressive conditional heteroskedasticity; GARCH-M: generalized autoregressive conditional heteroskedasticity in mean; P_h : historical wind power; P : predicted wind power; RBF: radial basis function; V_h : historical wind speed; V : predicted wind speed.

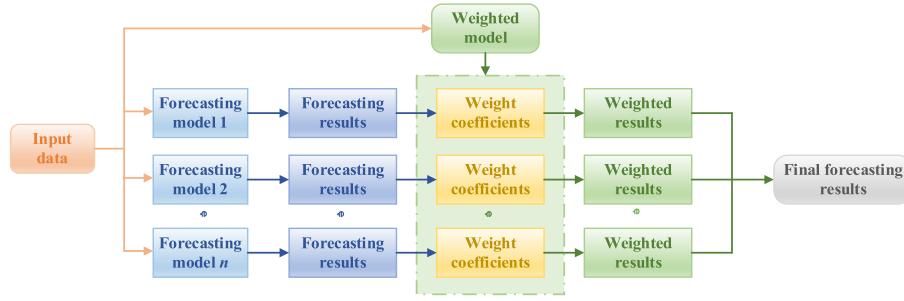


Fig. 6. Flowchart of horizontal combination methods.

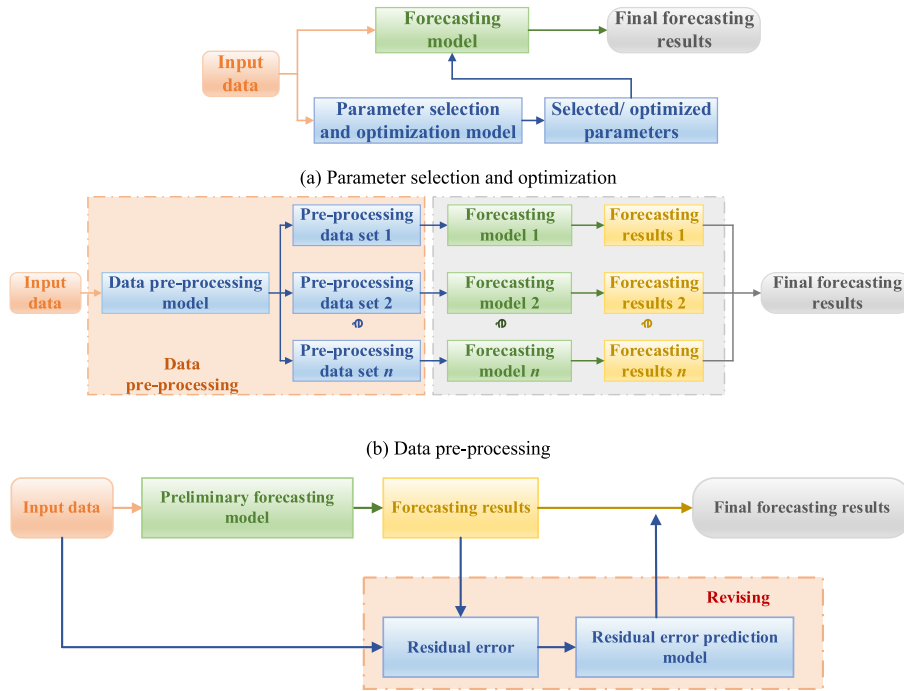


Fig. 7. Flowchart of three vertical combination approaches.

caused by wave, tsunami, storm, etc., and could not be maintained in time due to remote distance. Hence, it is arduous to collect observational data for wind prediction from ocean;

- From the perspective of communication connection. Offshore wind farms are usually connected to onshore control centers through submarine cables with tens of kilometers or wireless transmission. Consequently, distortion and packet loss of data are more likely to occur than onshore wind farm, which usually decrease the accuracy of wind forecasting;

- From the perspective of forecasting technologies. Physical models of offshore wind farm must carefully consider critical factors of ocean conditions, such as high specific heat capacity of sea, heat effect of wind current, and amplified wake effect, etc. (James et al., 2018). While statistical methods often require more diverse and larger-scale input data including historical wind speed and direction data, geographic data, NWP data, and real-time monitoring, to train models to adapt severe surroundings. Therefore, development of a forecasting approach with strong robustness, high reliability, and low data dependency is highly valuable for offshore wind farms.

Table 8 gives a summary of six offshore forecasting methods in the last decade. Work (James et al., 2018) presents a high-resolution

rapid updating NWP model considering complex ocean environment, which can realize 80 m-height wind prediction with 1 h ahead. Statistical methods such as GP (Charhate et al., 2009), SVM (Dangar et al., 2010), recurrent NNs (Balluff et al., 2015) are also adopted, while their scale of input data collection are 36 months, 16 months, and 15 years, respectively. Meanwhile, DWT is employed to preprocess wind speed and NWP data of complex ocean surrounding (Panapakidis et al., 2019). Meanwhile, study (Lin et al., 2020) utilizes wind speed, nacelle orientation, yaw error, blade pitch angle, and ambient temperature as input features of deep learning model to predict high-fluctuating offshore wind power.

4. Uncertainty forecasting

Wind speed and power forecasting is also known as deterministic/point prediction, which is inevitably influenced by uncertain factors, such as wind speed randomness, observation precision, and status of wind turbines (Yan et al., 2015). Thus, uncertainty forecasting has drawn ever-increasing attentions to decrease decision risk to power system.

Table 6

Chronological summary of forty-one combined methods for wind speed and power prediction.

Approach	Year	Horizontal combination	Vertical combination			Inputs	Time-scale	Space-scale	Forecast variables	Metrics	Accuracy improvement	Features
			Parameter selection	Data pre-processing	Data post-processing							
Maximum entropy, NNs (Han and Liu, 2010)	2010	✓				V_h	Short	Single wind farm	V	RMSE	21.92% RMSE improvement over persistent (6 h ahead).	● Maximum entropy is firstly introduced to select combination coefficients.
LS-SVM, covariance optimization, equal weight average (Zhao et al., 2011b)	2011	✓				V_h	Medium	Single wind farm	V	MAE, MRSE	More than 8% MAE improvement over LS-SVM.	● Adaptive; ● Time-varying weight coefficient method.
DWT, NNs (Catalao et al., 2011)	2011		✓			P_h	Short	Single wind farm	P	MAPE, SDE, SSE	4.00% MAPE improvement over single NN.	● Computation time less than 10 s.
ANN, ASD (Cui et al., 2012)	2012		✓			P_h	Short	Single turbine	P	NMAE, NRMSE	NMAE:13.10%; NRMSE:16.53%.	● Sparser decomposition results.
ARIMA, ELM, EWT, GPR, LS-SVM (Wang and Hu, 2015)	2015	✓	✓			V_h	Short	Single wind farm	V	MAE, MAPE, RMSE	18.34% MAPE improvement over EWT-AR.	● GPR integrates individual forecasts nonlinearly.
Elman recurrent NN, SIA, SVR (Wang et al., 2015)	2015		✓			V_h	Long	Single wind farm	V	MAE, MAPE, MPE, MSE	3.01% MAPE improvement over single NN.	● Outliers in data handled; ● Seasonal effect from data extracted.
ESNs, GA, PCA, SC, WD (Liu et al., 2015)	2015		✓	✓		V_h	Short	Single wind farm	V	MAE, MAPE, NRMSE	15.14% MAPE improvement over ESNs.	● Number of input variables reduced.
PSO, PCA, Reduced SVM (Kong et al., 2015)	2015		✓			V_h , NWP data	Short	Single wind farm	V	MAE, MAPE, RMSE	5.65% MAPE improvement over RBF.	● Manual parameter tuning avoided; ● Reduced dimension of observed variables.
ANNs, CSO, WPD (Meng et al., 2016)	2016		✓	✓		V_h	Short	Single wind farm	V	MAE, MAPE, RMSE	91.08% MAPE improvement over single NN.	● Premature of NNs solved.
BPNN, EEMD, GA (Wang et al., 2016)	2016		✓	✓		V_h	Very short, Short	Single wind farm	V	MAPE, RMSE	59.57% MAPE improvement over BPNN-GA.	● Improved accuracy and computational efficiency.
GA, PSR, SVR (Santamaría-Bonfil et al., 2016)	2016		✓	✓		V_h	Short	Single wind farm	P	MAE, MASE, RMSE	Average of 9.5% improvement over ARIMA.	● Feasibility of PSR is verified by a systematic analysis over wind speed.
ITD, LS-SVM (Zhang et al., 2016)	2016			✓		P_h	Short	Single wind farm	P	MAPE, NMAXAE, NRMSE	More than 10% improvement by introduced ITD.	● Satisfactory accuracy and stability.
AdaBoost-ELM, CEEMDAN, VMD (Peng et al., 2017a)	2017			✓		V_h	Short	Single wind farm	V	MAE, MASE, RMSE	24.24% RMSE improvement over AdaBoost-ELM.	● Two-stage decomposition algorithm containing VMD and CEEMDAN.
ANNs, CEEMDAN, MFPA (Zhang et al., 2017a)	2017	✓	✓	✓		V_h	Short	Single turbine	V	MAE, MAPE, RMSE, SSE	66.23% MAPE improvement over single NN.	● No negative constraint theory based; ● Optimal weights of model are determined by MFPA.
Bayesian KF, NWP model (Galanis et al., 2017)	2017				✓	NWP data	Short	Single wind farm	V	Bias, Nash-Sutcliffe efficiency, RMSE	45.33% RMSE improvement over single NWP model (case 1).	● Systematic prediction error is eliminated in initial NWP model; ● Reduced uncertainty of error.
CEEMD, Elman NN, MOWOA (Wang et al., 2017b)	2017		✓	✓		V_h	Very short	Single wind farm	P	DC, IA, MAE, MAPE, MdAPE NMSE, r , RMSE	53.36% MAE improvement over Elman NN.	● Prominent accuracy and stability.
GRNN, modified BA, SSA (Xiao et al., 2017)	2017		✓	✓		V_h	Medium	Single turbine	V	MAE, MAPE, MSE	10% overall improvement over GRNN.	● Feasibility is verified by abundant experiments.
mRMR, weighted regular ELM (Xing et al., 2017)	2017		✓			V_h , NWP data	Short		V	MAPE, RMSE	/	● Improved generalization and fitting ability.

(continued on next page)

Table 6 (continued)

Approach	Year	Horizontal combination	Vertical combination			Inputs	Time-scale	Space-scale	Forecast variables	Metrics	Accuracy improvement	Features
			Parameter selection	Data pre-processing	Data post-processing							
PSR, KPCA, CVR, COR (Sun et al., 2017)	2017		✓	✓		V_h , NWP data	Long	Single wind farm	V	MAE, MAPE, RMSE, DC	15.02% MAPE improvement over PCA.	● Low computation time; ● Well robustness.
AdaBoost.RT, BFGSNNs, SE, VMD, WD, WF (Liu et al., 2018b)	2018			✓	✓	V_h	Short	Single wind farm	V	MAE, MAPE, RMSE	38.19% MAPE improvement over original NN (site 1, 5-step).	● Large multi-step wind speed prediction; ● Secondary decomposition combining WD and VMD.
ANNs, GWO, ICEEMDAN (Song et al., 2018)	2018	✓	✓	✓		V_h	Medium	Single wind farm	V	MAE, MAPE, RMSE, SSE	41.24% MAPE improvement over EMD (site 1).	● Reduced high-frequency noise.
DBN, k -means clustering (Wang et al., 2018)	2018			✓		V_h	Short	Single wind farm	P	MAPE, NMAE	68.97% MAPE improvement over BP (spring).	● Historical samples are extracted to enhance accuracy.
EEMD, KFCM, Wavelet NN (He et al., 2018)	2018			✓		V_h	Medium	Single wind farm	V	MAE, MAPE, MSE	68.69% MAPE improvement over wavelet NN.	● KFCM considers data similarity; ● Reduced requirements of training models.
EMD-ARIMA, EMD-BA-BPNN, EMD-BA-ENN, SVR (Li et al., 2018)	2018	✓	✓	✓		V_h	Medium	Single turbine	V	MAE, MAPE, MSE	23.21% MAPE improvement over EMD-ARIMA.	● Variable weights based; ● Speed data prediction of nonlinear and linear characteristics.
FEEMD, improved BPNN, PSR, SE (Sun and Wang, 2018)	2018			✓		V_h	Medium	Single wind farm	V	MAE, MAPE, RMSE	51.10% MAPE improvement over improved BPNN.	● High computing speed; ● Strong generalization; ● Elegant nonlinear mapping.
CEEMD, CMI, ELM (Huang et al., 2018)	2018			✓		V_h	Short	Single turbine	V	MAPE, RMSE	3.86% MAPE improvement over MI.	● Low redundancy feature selection prediction method.
PACF, QRF, meta learning strategy (Hu et al., 2018)	2018	✓		✓		V_h	Short	Single wind farm	P	NMSE, NRMSE, NMAE	29.30% NMSE improvement over quantile RNN (data A).	● Satisfactory prediction accuracy.
WTD, RNN, ANFIS (Cheng et al., 2018)	2018			✓		V_h	Very short	Single turbine	V	MAE, RMSE	29.55% RMSE improvement over ANN (1-step).	● Adequate to both deterministic and probabilistic prediction.
EWT, Elman NN, LSTM (Liu et al., 2018a)	2018			✓		V_h	Short	Single wind farm	V	MAE, MAPE, RMSE	51.44% MAPE improvement over Elman NN (1-step).	● LSTM and Elman NN are utilized to predict low-frequency and high-frequency sub-layer of wind speed, respectively.
ARIMA, CEEMDAN, MOGOA, NNs (Niu and Wang, 2019)	2019	✓	✓	✓		V_h	Medium	Single wind farm	V	MAE, MAPE, MSE, SSE	12.29% MAPE improvement over CEEMDAN-ARIMA (site 1, 1-step).	● Reduced high-frequency noise in data.
ARMA, PCA, RBF, VMD (Zhang et al., 2020)	2019			✓	✓	V_h , NWP data	Short	Single wind farm	V	MAE, MAPE, MSE, rMBE	5.56% MAPE improvement by data-post processing (Spain data).	● Adequate to deterministic prediction and uncertainty prediction; ● Satisfactory reliability.
Convolutional SVM, EMD, SSA (Mi et al., 2019)	2019			✓		V_h	Short	Single wind farm	V	MAE, MAPE, RMSE	28.48% MAPE improvement over convolutional SVM (experiment 1).	● Enhanced performance in short-term multi-step wind speed forecasting; ● Requirement of high-quality data.
Data-driven (Yan and Ouyang, 2019)	2019			✓		P_h , V_h	Long	Single wind farm	P	MAE, RMSE	69.50% MAE improvement by data-post processing.	● Proper precision; ● Low cost.
DBM, EEMD (Santhosh et al., 2019)	2019			✓		V_h	Short	Single wind farm	V	MAE, MAPEM, RMSE	77.33% MAPE improvement over DBM.	● Easy to develop; ● High precision; ● Robust hybrid deep learning strategy.
GRUNNs, NWP model (Ding et al., 2019)	2019			✓		NWP data	Short, Medium, Long	Single wind farm	P	MAE, RMSE	63.32% RMSE improvement over NWP model (22 days).	● Weight time series are presented.
	2019		✓	✓		P_h	Medium		P	MAE, MAPE, RMSE		● Robust hybrid model; ● Proper accuracy and stability.

ICEEMDAN, MOMFO, wavelet NN (Du et al., 2019)	2020 ✓	✓	V_h , NWP data, Real-time monitoring data	Short	Single wind farm	MAPE, NMSE	62.38% MAPE improvement over wavelet NN (dataset 1, 1-step). 15.75% MAPE improvement over SVM.	● Increased difference of input samples among base learners; ● High precision and reliability.
Bayesian model, SOM clustering, BPNN, RBF and SVM (Wang et al., 2020b)	2020	✓	V_h , NWP data	Short	Single wind farm	MAPE, NMAE, NRMSE, R^2	21.40% MAPE improvement over BPNN.	● Performance analyzed synthetically; ● Prominent precision.
IDA, SVM (Li et al., 2020d)	2020	✓	NWP data	Medium	Single wind farm	RMSE	49.66% RMSE improvement over SVM.	● Decreased unreliability of prediction; ● SSO firstly utilized to optimize FCM.
NWP model, SSO, FCM clustering, Apriori laws (Li et al., 2020c)	2020	✓	V_h	Short	Single wind farm	MAE, RMSE	More than 7% RMSE improvement over ANN (1-step).	● Wind turbulent component considered; ● Adequate to both deterministic and probabilistic prediction.
CNN, ACF (Zhao et al., 2020)	2020	✓	V_h	Very short	Single wind farm	MAE, MAPE, RMSE	50.21% MAPE improvement over neural network	● Dynamic characteristics of wind speed captured.
SSA, neural network structural learning (Mi and Zhao, 2020)	2020	✓			Single wind farm		structural learning (test 1).	● Robust and high-precision.

***Note.** ANN: artificial NN; ANFIS: adaptive neuro fuzzy inference system; BFGSNNs: Broyden–Fletcher–Goldfarb–Shanno NNs; BPNN: back-propagation NN; COR: competition over resource; CSO: crisscross optimization algorithm; CVR: core vector regression; DBM: deep Boltzmann machine; DBN: deep belief network; ELM: extreme learning machine; ESNS: echo state networks; IDA: improved dragonfly algorithm; GPR: Gaussian process regression; GRNN: general regression NN; GRUNNs: bidirectional gated recurrent unit NNs; GWO: grey wolf optimizer; LS-SVM: least-squares SVM; MOMFO: multi-objective moth-flame optimization; MOMOA: multi-objective whale optimization algorithm; P_h : historical wind power; P : predicted wind power; QRF: quantile random forest; RBF: radial basis function; SSO: shark small optimization; SVR: support vector regression; V_h : historical wind speed; V : predicted wind speed; WF: wavelet filter.

4.1. Probabilistic forecasting

Probabilistic forecasting focus on when and what degree forecasting error happens, which possesses several different forms (Zhang et al., 2019), e.g., probability density function (PDF), cumulative distribution function (CDF), discrete probabilities, quantiles, intervals, mean, etc. In general, they can be categorized into two groups: parametric approaches and non-parametric approaches. The former ones are based on assumption that probabilistic outcomes follow pre-defined distribution, such as generalized logit-normal (Pinson, 2012), Beta (Zhang et al., 2013), Gaussian (Kou et al., 2014), and versatile distributions (Zhang et al., 2013), then other technologies are applied to determine parameters. Obviously, the largest limitation is that sequences are difficult to completely obey pre-defined shape. Meanwhile, the latter ones are distribution-free but need more computation resources, which include adaptive robust multi-kernel regression (ARMKR) (Wang et al., 2017a), adapted resampling (Pinson and Kariniotakis, 2010), kernel density estimator (Zhang and Wang, 2016) and quantile regression (Wang et al., 2020c), etc.

4.2. Risk index forecasting

Risk index forecasting provides intuitive uncertainty information to operators which can be denoted as discrete indicators, such as color codes (i.e., red-green-yellow) and number values (i.e., from 1 to 5) (Yan et al., 2015), as well as risk indices including Meteo-Risk Index (MRI) (Pinson and Kariniotakis, 2004), Normalized Prediction Risk Index (NPRI) (Pinson et al., 2009b), and Predict at Risk Index (PaR) (Yan et al., 2013).

4.3. Scenario forecasting

Scenario forecasting offers a series of power generation circumstances in each moment, which can describe uncertainty and effectively deal with multi-period and multi-stage decision-making. Common scenarios forecasting include Monte Carlo algorithm (Pinson et al., 2009a), multivariate Gaussian random variable (Pinson et al., 2009a), and multivariate ARMA (Morales et al., 2010). Moreover, clustering method (Sumaili et al., 2011) and probability distance based backward reduction (Sharma et al., 2013) are proposed to reduce scenario number to balance amount and precision.

Here, Table 9 summarizes fourteen methods of uncertainty forecasting methods in the last decade. In general, non-parametric approaches of probabilistic forecasting are distribution-free which can achieve short and medium-term forecasting. While parametric approaches have lower forecasting accuracy due to wind series may not obey pre-defined distribution. Moreover, evaluation criteria of risk index forecasting are insufficient which limit their applications. Scenario forecasting needs numerous historical data to generate a series of scenarios, hence scenario reducing technologies are promising.

5. Ramp events forecasting

Sharp boost or drop of wind power caused by severe circumstances are known as ramp events, while such abnormal states might result in active power imbalance in power system or even large-scale load shedding (Cristobal et al., 2015). Besides, definition of ramp events is also controversial in selecting time interval or power threshold (Ferreira et al., 2010).

5.1. Deterministic forecasting

Deterministic forecasting methods directly capture ramp events

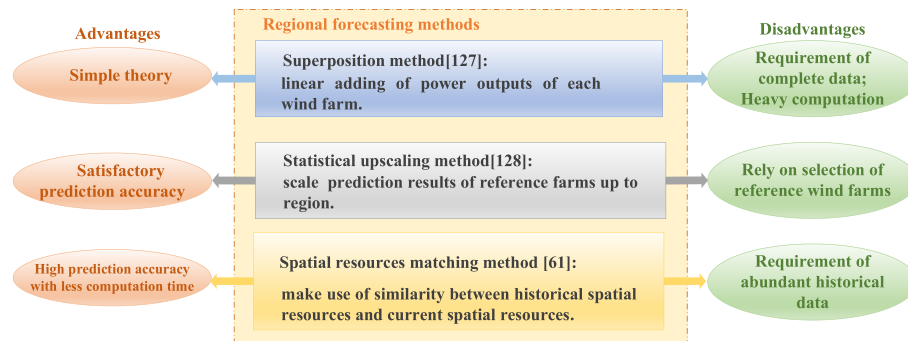


Fig. 8. Comparison of three regional forecasting methods.

Table 7

Chronological summary of four spatial correlation forecasting methods and six regional forecasting methods.

Approach			Year	Inputs	Time-scale	Space-scale	Forecast variables	Metrics	Features		
Spatial correlation	Physical method	Physical approach (Kong et al., 2015)	2017	V_h , Geographic data	Short	Single wind farm	V	NRMSE	● Correlation among wind turbines described; ● Wind uncertainty analyzed.		
	Statistical methods	Fuzzy model, GA (Damousis et al., 2004)	2004	V_h , D_h	Short	Single wind farm	P , V	RMS	● Significant improvement compared with Persistent method.		
		Artificial NNs (Velazquez et al., 2011)	2011	V_h , D_h	Long	Single turbine	P , V	MARE, r	● Angular wind direction employed as inputs.		
		Measure-correlate-predict (Ali et al., 2018)	2018	V_h	Long	Single turbine	V	Bias, MAE, MSE, R^2 , RMSE, SDE, SDbias	● Evaluation of wind potential.		
Regional forecasting	Statistical upscaling methods	Upscaling approach (Ishihara et al., 2007)	2007	P_h , Geographic data	Long	Region	P	RMSE	● Error decreases with increasing number of reference wind farms.		
		Upscaling approach, data mining (Ozkan and Karagoz, 2019)	2019	P_h , V_h , NWP data	Short	Region	P	NMAE, NRMSE	● Bottom-up approach; ● Evaluation of wind resource.		
	SRMA	Smoothing techniques (Santamaría-Bonfil et al., 2016)	2012	V_h , Geographic data	Short	Region	P	NMAE, NRMSE	● Non-parametric; ● Rapid calculation.		
		Modified SRMA (Peng et al., 2017b)	2017	P_h , V_h , Geographic data	Short	Region	P	RMSE	● Improved prediction accuracy with less computation burden.		
		Gaussian processes (Xue et al., 2020)	2020	P_h , NWP data	Medium	Region	P	NMAE, NRMSE	● Adequate to both deterministic and probabilistic prediction.		
		GWO, MSVM (Lu et al., 2020)	2020	P_h	Long	Region	P	DC, NMAE, NRMSE	● Novel spatio-temporal prediction framework.		

*Note. D_h : historical wind direction; GA: genetic algorithm; GWO: grey wolf optimizer; MSVM: multi-output support vector machine; P_h : historical wind power; P : predicted wind power; SRMA: Spatial resources matching approach; V_h : historical wind speed; V : predicted wind speed.

Table 8

Chronological summary of six offshore forecasting methods.

Approach		Year	Inputs	Time-scale	Space-scale	Forecast variables	Metrics	Features
Physical approaches	High-resolution rapidly updating NWP model (James et al., 2018)	2018	NWP data	Short	Region	V	Bias, RMS, RMSE	● 80 m wind speed estimation in coastal regions.
Statistical approaches	GP (Charhate et al., 2009)	2009	V_h , D_h	Short	Single turbine	V , D	MAE, r , RMSE	● Wind speed and direction predicted simultaneously.
	SVM (Dangar et al., 2010)	2010	V_h , D_h	Short	Single wind farm	V	MSE	● Robust tool for data mining.
	Recurrent NNs (Balluff et al., 2015)	2015	V_h , D_h , Geographic data, NWP data	Medium	Single wind farm	V	/	● Helpful to decision-making problem.
	ANFIS, FFNN, DWT (Panapakidis et al., 2019)	2019	V_h , D_h , NWP data	Very short	Single wind farm	V	AE, MAE, MARNE, RMSE	● Simple structure;
	Deep learning, IF (Lin et al., 2020)	2020	Real-time monitoring data	Long	Single turbine	P	MSE	● High-efficient model.
								● Powerful anomaly detection tool.

*Note. ANFIS: adaptive neuro fuzzy inference system; D_h : historical wind direction; D : predicted wind direction; FFNN: feed-forward neural network; GP: Genetic programming; IF: isolate forest; P : predicted wind power; V_h : historical wind speed; V : predicted wind speed.

Table 9
Chronological summary of fourteen uncertainty forecasting methods.

Approach			Year	Inputs	Time-scale	Space-scale	Forecast variables	Metrics	Features
Probabilistic forecasting	Parametric approaches	Generalized logit–normal distributions (Pinson, 2012)	2012	P_h, D_h	Very short	Single wind farm	PDF	CRPS, MADC, NMAE, NRMSE	● Nonlinear and double-bounded nature of wind power described.
		Versatile distributions (Zhang et al., 2013)	2013	P_h	Very short	Single wind farm	PDF, CDF	Computation time	● Wind power prediction errors determined for any forecasting times-term and magnitude.
		Warped Gaussian process (Kou et al., 2014)	2014	V_h , NWP data	Very short, Short	Single wind farm, Region	Intervals	NLPD	● Adequate to both interval and point forecasting; ● Time adaptive.
	Non-parametric approaches	Adapted resampling (Pinson and Kariniotakis, 2010)	2010	V_h	Short	Single wind farm	Intervals	Skill score	● Fuzzy inference mode; ● Low computation cost.
		Kernel density estimator (Zhang and Wang, 2016)	2016	V_h, D_h	Medium	Single wind farm	Intervals, quantiles	QS, RMSE	● Adequate to both probabilistic and point forecasting.
		ARMKR (Wang et al., 2017a)	2017	V_h	Medium	Single wind farm	PDF	ACE, CWC, PICP, PINAW, WS	● High fitting ability for continuous complex distributions; ● Requirement of high-resolution data.
		Quantile regression (Wang et al., 2020c)	2020	V_h, D_h , NWP data	Short	Region	Intervals	Pinball, reliability, sharpness	● Novel regime-switching strategy; ● Dimension disaster avoided.
	Risk index forecasting	MRI (Pinson and Kariniotakis, 2004)	2004	NWP data	Short	Single wind farm	/	/	● Fine-tune of confidence intervals achieved; ● Probability of outliers occurring provided.
		NPRI (Pinson et al., 2009b)	2009	V_h, D_h	Short, Medium	Single wind farm	/	/	● Energy imbalance level described.
		PaR (Yan et al., 2013)	2013	P_h , NWP data	Short	Single wind farm	/	/	● Fluctuation range of predictive power under different confidence level provided.
Scenario forecasting		Multivariate Gaussian variable, covariance matrix (Pinson et al., 2009a)	2009	P_h	Short	Single wind farm	Wind power scenarios	Standard deviation, skewness	● Predictive distributions and interdependence structure of prediction errors considered.
		Multivariate ARMA, covariance matrix (Morales et al., 2010)	2010	P_h	Short	Region	Wind speed scenarios	Mean and standard deviation	● Marginal distribution, temporal, and spatial correlations of wind speed incorporated.
		Clustering techniques (Sumaili et al., 2011)	2011	P_h	Medium	Single wind farm	Wind power scenarios	Maximum deviation	● Reduced calculation burden.
		ARMA, probability distance based backward reduction (Sharma et al., 2013)	2013	V_h	Short	Single wind farm	Wind power scenarios	Mean and standard deviation	● Reduced number of scenarios with minimal variation in deviation.

***Note.** ACE: average coverage error; CRPS: continuous rank probability score; CWC: coverage width-based criterion; D_h : historical wind direction; MADC: mean absolute deviation from calibration; NLPD: negative log predictive density; P_h : historical wind power; PICP: prediction interval coverage probability; PINAW: prediction interval average width; QS: quantile score; V_h : historical wind speed; WS: Winkler score.

based on characteristic variables, such as ramp rates, magnitude, and ramp duration (Cristobal et al., 2015). Typical methods include data-mining (Zheng and Kusiak, 2009), regime-switching ANNs (RS-ANNs) (Gallego et al., 2011), stream-based hmm ramp event detector (SHRED) (Ferreira et al., 2012), optimized swinging door algorithm (OpSDA) (Zhang et al., 2017b), and Markov-switching-auto-regression model (MSAR) (Ouyang et al., 2019).

5.2. Probabilistic forecasting

Probabilistic forecasting focuses on probability distribution of characteristic variables. Thus far, major methods, such as Monte Carlo algorithm (Ferreira et al., 2013), NNs (Cui et al., 2015), generalized Gaussian mixture model (GGMM) (Cui et al., 2017), and data-driven (Cao et al., 2019) are developed based on generated scenarios. Moreover, multi-threshold conditional autoregressive multinomial logit (CARML) model (Taylor, 2017) and time series similarity search algorithm (Cao et al., 2018) are direct forecasting methods based on wind prediction errors.

Table 10 summarizes eleven ramp events forecasting methods. In particular, probabilistic forecasting of ramp events is more significant since wind speed/power climbing is a small-probability event. Besides, deterministic forecasting usually ignores physical properties, which are vital factors causing ramp events and is of great importance for future study.

***Note.** ACS: average coverage error; AIS: average interval score; P_h : historical wind power; PICP: prediction interval coverage probability; PINC: prediction interval nominal confidence; V_h : historical wind speed.

6. Discussions

Numerous existing approaches on wind forecasting have been reviewed in the last three parts, which are critical to wind power penetration improvement (Chen et al., 2019), wind energy consumption promoting (Liu et al., 2020), and power grids stability

enhancement (Mensou et al., 2020). Besides, it is difficult and incorrect to compare various methods only based on the aforementioned evaluation criteria due to each method is carried out under customized simulation cases (Jung and Broadwater, 2014). Thus, this section tends to focus on advantages/contributions, disadvantages/limitations, applicability, data size, and accuracy (Liu and Chen, 2019) of each model for a more practical comparison, as demonstrated in Table 11. To this end, a comprehensive discussion/analysis of all wind forecasting methods based on three perspectives (wind speed and power forecasting, uncertainty forecasting, and ramp events forecasting) are summarized as follows:

- Conventional wind speed and power forecasting approaches focus on single-valued prediction of onshore single wind turbine or single wind farm based on local data, which consist of three sub-classes:
 - (i) Physical methods utilize super computers to solve complex kinematic physical equations, which inputs are weather data such as temperature, relative humidity, and atmospheric pressure. These approaches just operate once or twice a day because of the difficulty to obtain sufficient weather data in a short period of time and high computation costs, such that only medium and long-term forecasting are achievable (corresponding to Section 3.2.1);
 - (ii) Classical statistical methods (time series analysis and KF) with the merits of simple structure and low costs fill the gaps in very short and short-term forecasting. These models are linear predictors based on abundant historical data, while wind speed series have highly nonlinear features. Hence, conventional statistical approaches cannot realize a consistent performance in various operation conditions, which are merely employed as reference models. Machine learning approaches describing nonlinear mapping between historical and forecasted wind, which have become the mainstream in the last decade and can be used in all time-term.

Table 10
Chronological summary of eleven ramp events forecasting methods.

Approach	Year	Inputs	Time-scale	Space-scale	Forecast variables	Metrics	Features
Deterministic forecasting	Data-mining (Zheng and Kusiak, 2009)	2009 P_h, V_h	Very short, Short	Single wind farm	Wind power rates	MAE, standard deviation	● Decreased dimension of inputs.
	RS-ANN (Gallego et al., 2011)	2011 P_h	Short	Single wind farm	Wind power local gradient	Improvement over Persistence	● Effective for detecting ramp-up, ramp-down, and no-ramp state.
	SHRED (Ferreira et al., 2012)	2012 P_h	Short	Single wind farm	Magnitude, direction and duration	Misclassification costs, true skill score	● Descriptive and cost-effective model; ● Ramp events effectively predicted.
	OpSDA (Zhang et al., 2017b)	2017 P_h	Short	Single wind farm, Region	Magnitude, direction and duration	/	● Four different ramp definitions provided.
	MSAR, improved SDA (Ouyang et al., 2019)	2019 P_h, V_h	Medium	Single wind farm	Duration	MAE, RMSE, correlation coefficient	● Residual correction considered to improve performance.
Probabilistic forecasting	Monte Carlo algorithm (Ferreira et al., 2013)	2013 P_h	Short	Single wind farm	Ramp probabilities	Brier score, critical success index	● Based on PDF; ● Strong robustness.
	NN model, genetic algorithm (Cui et al., 2015)	2015 P_h	Short	Single wind farm	Ramp probabilities	/	● Wind power generation stochastic process effectively approximated.
	Multi-threshold CARML (Taylor, 2017)	2017 P_h	Short	Single wind farm	Ramp probabilities	Brier score	● Ramp events probabilities with more than one threshold estimated simultaneously.
	GGMM, OpSDA, data-driven (Cui et al., 2017)	2017 P_h	Very short	Single wind farm	Probability of start time	AIS, ACE, PICP, PINC, interval score	● Excellent ability of fitting distributions.
	Time series similarity search algorithm (Cao et al., 2018)	2018 P_h, V_h	Short	Single wind farm	Ramp probabilities	/	● Ramp events directly predicted from wind speed forecasts.
	Data-driven probability inequality (Cao et al., 2019)	2019 P_h, V_h	Short	Single wind farm	Ramp probabilities	Mean and standard deviation	● Computationally tractable convex optimization; ● No need for exact distributions.

Table 11
Comparison of three categories of wind forecasting approaches.

Category		Advantages/contributions		Disadvantages/limitations	Purpose	Applicability	Usage frequency	Implementation difficulty	Data size	Accuracy
Wind speed and power forecasting	Physical approaches		● Mature methods; ● No need for historical data.	● Sensitive to initial conditions; ● Rely on stable atmospheric conditions; ● Heavy calculation burden.	● Single-valued prediction of wind speed or power in expected time.	● Medium-term and long-term forecasting. ● Stable weather.	**	**	***	**
	Statistical approaches	Time series analysis	● Complicated physical process ignored; ● Simple structure.	● Numerous historical data; ● Poor performance on low-order model; ● Many parameters in high-order model.		● Short-term forecasting; ● Relatively stable time sequences.	***	**	***	**
		KF	● Optimal linear recursive estimation.	● Numerous historical data; ● Difficult to establish state equation and measurement equation.		● Very short-term and short-term forecasting.	**	**	***	**
		NNs	● Self-adaptive; ● Self-learning; ● NO need for accurate system model; ● Stochastic nature properly handled.	● Numerous historical data; ● Model over-fitting; ● Poor generalization; ● Rely on selected parameters.		● All time-term; ● Large-sized samples.	***	***	*****	***
		SVMs	● Less historical data than NNs; ● Less calculation burden than NNs;	● Low robustness; ● Rely on selected parameters.		● All time-term; ● Medium-sized samples.	***	***	****	***
	Combined approaches	Horizontal combination	● Dimension disaster avoided. ● Reduced risks during extreme events. ● Easy to implement.	● Requirement of extra weighted model; ● Heavy computation burden.		● All time-term; ● In pursuit of robustness.	****	*****	*****	*****
		Vertical combination	● Manual parameter tuning avoided.	● Hard to code.		● All time-term. ● Combined models including NNs and SVMs. ● All time-term.	*****	****	****	****
		Parameter selection	● High precision; ● Robustness.	● Complicated structure; ● Long computation time.		● Numerous unstable input data. ● All time-term; ● In pursuit of high accuracy.	*****	****	*****	*****
		Data pre-processing	● Help for reducing systematic error.	● Long computation time.			***	****	*****	*****
	Spatial correlation forecasting	Physical method	● Information about physical phenomena and meteorological condition utilized.	● High complexity and cost due to time delay between different sites.	● Wind resource evaluated at site where information is unavailable but neighbouring stations are available.	● Meteorological stations available in neighbouring area.		*****	*****	**
		Statistical methods	● Pattern based; ● Complicated physical process ignored.	● Requirement of large historical data; ● Dimension disaster.		● Wind farms available in neighbouring area.	**	****	*****	***

(continued on next page)

Table 11 (continued)

Category		Advantages/contributions		Disadvantages/limitations	Purpose	Applicability	Usage frequency	Implementation difficulty	Data size	Accuracy
Regional forecasting	Superposition method	● Simple theory.		● Requirement of complete data.	● Aggregated power output predicted in a region.	● Small-sized cluster.	^a	^a	****	^a
	Statistical upscaling methods	● Improved precision; ● Only data of reference wind farms is needed to scale results up to clusters.		● Difficult to determine reference wind farms.		● Medium-sized cluster.	***	***	**	**
	SRMA	● Easy to implement; ● High prediction accuracy.		● Requirement of abundant historical data.		● Large-sized cluster.	**	**	***	***
Offshore forecasting	Physical approaches	● No need for historical data.		● Sensitive to initial conditions; ● Heavy calculation burden. ● Additional factors of ocean condition considered.	● Single-valued prediction of wind speed or power in expected time of offshore area.	● Mostly in medium-term and long-term forecasting.	^a	***	****	**
	Statistical approaches	● Pattern based; ● Complicated physical process ignored.		● Requirement of large historical data.		● Mostly in very short-term and short-term forecasting.	***	****	*****	*****
Uncertainty forecasting	Probabilistic forecasting	Parametric	● Simple; ● Low calculation cost. ● Distribution-free.	● Rely on pre-defined distribution.	● When and what degree forecasting error happens represented.	● Very short-term forecasting.	^a	**	**	**
		Non-parametric		● Requirement of abundant historical data; ● High calculation cost.		● Short-term and medium-term forecasting.	****	***	****	****
	Risk index forecasting	● User-oriented representation.		● Difficult to evaluate performance.	● Intuitive and simple uncertainty information offered.	● Risk quantified; ● Security evaluation.	**	***	****	***
Ramp events forecasting	Scenario forecasting	● Development of prediction uncertainty as time goes on described.		● Model redundancy problem; ● Heavy calculation burden.	● Decision-making problems solved.	● Trading strategy; ● Unit commitment; ● Economic dispatching.	***	****	*****	*****
	Deterministic forecasting	● Intuitive description.		● Lack of uniform definition of ramp events; ● Physical properties ignored; ● Only appropriate for specific case.	● Characteristic variables described quantitatively.	● Ramp rates, magnitude, and ramp duration predicted.	**	***	****	***
	Probabilistic forecasting	● More risk information provided.		● Lack of uniform definition of ramp events; ● Only appropriate for specific case; ● Lack of available samples.	● Ramp warning, events	● Probability distribution of ramp events predicted.	***	****	*****	***

^a Note. *: very low/short; **: low/short; ***: medium; ****: high/long; *****: very high/long.

- Specifically, NNs are self-adaptive and self-learning, but have disadvantages of model over-fitting, poor generalization, and high computation burdens. SVMs overcome above three shortcomings of NNs which can realize precise predictions for more general situations (corresponding to Section 3.2.2);
- (iii) The problem of machine learning approaches is that their forecasting results highly rely on the selected parameters. Fortunately, meta-heuristic algorithms owing to the strengths of global optimization and model-free, are adopted to determine parameters of NNs and SVMs. These methods belong to parameter selection and optimization approaches of vertical combination methods, which combine single prediction model with auxiliary methods in different forecasting links. Moreover, the other two auxiliary methods are data pre-processing and data post-processing approaches. Data pre-processing methods can realize a significant accuracy improvement via enhancing quality of input data. In particular, signal decomposition methods have the most highly usage frequency, which can decrease forecasting error up to 70% (He et al., 2018). However, dimensionality disaster is the critical defect of these methods, thus they are mostly used for medium or small-scale low fluctuation data. In order to handle numerous data, dimensionality deduction-based approaches including feature selection and feature extraction are recommended. Specially, feature selection methods are suitable for large-scale high relevance data with near 50% accuracy improvement (Li et al., 2020c), while feature extraction methods applied to large-scale high redundancy data with about 40% error reduction (Liu et al., 2015). Meanwhile, denoising-based approaches devote to removing noises of medium-scale drastic fluctuation data with about 50% accuracy improvement (Mi and Zhao, 2020). Moreover, data post-processing methods consider residual errors to revise preliminary prediction model which can further decrease more than 60% error (Ding et al., 2019). Note that no single method above is completely superior to others under various scenarios. In fact, different methods can be incorporated to establish more adequate models. The key issue is how to realize a proper trade-off between precision and efficiency. Lastly, horizontal combination approaches are another indispensable group, which own advantages of several prediction methods though an extra weighted model to reduce risks during extreme events. Hence, horizontal combination methods can realize a robust prediction, but require heavy computation burdens (corresponding to Section 3.2.3);
- Three novel trends of wind speed and power forecasting approaches have been developed in recent years, which are of great significance to the integration between wind energy and power systems:
 - (i) Spatial correlation forecasting is appropriate for wind resources evaluated at a site where information is unavailable but neighbouring stations are available. Different from conventional wind speed and power forecasting, physical methods of spatial correlation forecasting need to consider additional spatial topologies such as landform, surface roughness, wake effect, and arrangement of wind turbines, while statistical methods need more historical data from neighbouring wind farms. Spatial correlation forecasting often has high complexity and costs due to the measurements are based on multiple spatial correlated sites associated with time delays (corresponding to Section 3.3.1);
 - (ii) Regional forecasting provides whole wind power output in a region for market operators. Superposition method calculates the sum of wind power prediction from each wind farm, which is only used for small-sized cluster due to complex calculation. While statistical upscaling methods and SRMA can realize a faster and more accurate forecasting than superposition method according to spatial smoothing effects. Particularly, statistical upscaling methods utilize the data of reference wind farms to scale results up to medium-sized cluster, however it is relatively difficult to determine reference wind farms. SRMA does not need reference wind farms, while more historical data is required since it is designed for large-size cluster (corresponding to Section 3.3.2);
 - (iii) Offshore forecasting is still in the early phases of development. Physical models need to consider additional factors of ocean condition, such as high specific heat capacity of sea, heat effect of wind current, and amplified wake effect, etc. (James et al., 2018). Similarly, physical models of offshore forecasting are mostly used in medium-term and long-term forecasting. While statistical methods have good performance in very-short and short-term forecasting, which however require more diverse input data including historical wind speed and direction data, geographic data, NWP data, and real-time monitoring data. Besides, strong robustness, high reliability, and low data dependency are unique requirements of offshore forecasting due to harsh ocean environment (corresponding to Section 3.3.3);
 - Uncertainty forecasting indicates the occurrence time and degree of forecasting error, which can provide decision-makers more comprehensive information to maintain power system stability and improve wind farm capacity:
 - (i) Parametric approaches of probabilistic forecasting have less calculation burdens due to distribution shape only relies on several parameters which mostly aim at very short-term forecasting. However, unreasonable pre-defined distribution and variation of distribution shape with time might reduce the effectiveness of forecasting. Non-parametric approaches are distribution-free applied to short and medium-term forecasting, but have more input data and predicted values (densities or quantiles) which lead to high calculation costs (corresponding to Section 4.1);
 - (ii) Risk index forecasting represents the expected level of prediction error and intuitively reflects the operation of power system. However, their performance evaluation is difficult because the proposed indexes are customized in especial cases which lacks of generalizability (corresponding to Section 4.2);
 - (iii) Scenario forecasting dedicates to handling decision-making related to time such as trading strategy, unit commitment, and economic dispatching, etc., which need abundant wind data to generate scenarios. Hence, model redundancy issue and heavy calculation burden are their inherent drawbacks, and scenario reducing technologies are promising (corresponding to Section 4.3);
 - Although current power grid can effectively deal with small amounts of uncertainty, ramp events can still lead to considerable risk with increasing wind penetration. Hence, it is absolutely necessary to adopt suitable response strategies to predict ramp events accurately and timely:
 - (i) Deterministic forecasting of ramp events provides an intuitive description of ramp rates, magnitude, and ramp duration. However, there is still no consensus on the definition of ramp events, which indicates that specific method is only suitable for specific scenario. And physical properties should be considered in future study which are crucial factors causing ramp events (corresponding to Section 5.1);
 - (ii) Probabilistic forecasting estimates the probability distribution of ramp events. However, few studies focus on medium

and long-term forecasting in current literatures due to insufficient data samples of ramp events (corresponding to Section 5.2).

7. Conclusion

Due to its remarkable superiorities, e.g., pollution-free, high mobility and flexibility wind, and abundant reserves, wind energy has drawn widespread interests and researches. Note that accurate wind forecasting is of great significance to wind power systems, which is crucial for power dispatching and power grid security, as well as can bring considerable environmental, economic, and social benefits. Thus, there have been a series of prior reviews focus on summarizing these wind forecasting approaches. However, existing reviews on wind forecasting still have various limitations. For example, work (Jung and Broadwater, 2014) provide detailed classifications of wind prediction, but lack of the review of pre-processing methods. Literature (Tascikaraoglu and Uzunoglu, 2014) focuses on the review of combined methods on short-term wind prediction, thus some promising classifications are ignored. And reference (Hanifi et al., 2020) give a survey on time scale and accuracy improvement of wind power prediction, but the forecasting approaches and classifications are incomplete. In order to remedy the aforementioned shortcomings to offer readers comprehensive and systematic guidance for future researches in the related field, this work provides a state-of-the-art one-stop handbook of wind forecasting technologies from three perspectives, i.e., wind speed and power forecasting, uncertainty forecasting, and ramp events forecasting, while each category also includes several sub-categories. The main contributions can be concluded as follows:

- (a) Four classifications of wind data, along with twenty-seven data pre-processing approaches which can efficiently improve prediction precision are thoroughly covered;
- (b) Thirty-seven evaluation criteria of wind speed and power forecasting are carefully outlined for performance verification;
- (c) A total of one hundred and seven wind forecasting approaches are comprehensively summarized and compared based on their inputs, time-scale, space-scale, forecasting variables, metrics, and features, which aims to help readers utilize these approaches more effectively for future in-depth researches.

Moreover, eight perspectives and recommendations for further development in wind forecasting are proposed, as follows:

- (a) Continue to further study wind forecasting technologies is significant for facilitating wind energy penetration, promoting wind energy consumption, and maintaining power system stability;
- (b) Individual prediction models so far for conventional wind speed and power prediction are difficult to maintain a high accuracy under various scenarios, while combined prediction models are validated to effectively enhance prediction precision;
- (c) Spatial correlation forecasting is a powerful tool for potential wind farm site selection. Furthermore, data-driven approaches can be employed to solve such problem, which can considerably enhance data processing efficiency;
- (d) Regional forecasting is beneficial for formulating reasonable power generation plan, while selection of wind farms and cost reduction are worthy to be further studied;

- (e) Future wind farms are gradually moving from onshore land to offshore ones. But the installation and maintenance cost of offshore is higher than onshore. Hence, cost improvements could be achieved by studying new generators (e.g., floating turbines) and new materials;
- (f) The specific ocean conditions (e.g., high specific heat capacity of sea, heat effect of wind current, and amplified wake effect, etc.) of offshore forecasting can be considered to develop more precise forecasting models;
- (g) Combination of uncertainty forecasting and power system operation decision-making is significant, whose largest obstacle is scientific evaluation of these models which requires to be further investigated;
- (h) Ramp events forecasting can efficiently ensure a safe, stable, and economic operation of power system. However, definition of ramp events is controversial and forecast variables of ramp events are various, such that it is imperative to improve forecasting model adaptability.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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